VHDL Assignment 1 - NOT Gate

The purpose of this assignment is to implement a NOT gate in VHDL and verify its behavior using Logisim-evolution with the Questa VHDL simulator.

1. VHDL Source Code

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity not_gate is
Port (
a : in STD_LOGIC;
y : out STD_LOGIC
);
end not_gate;

architecture Behavioral of not_gate is
begin
y <= not a;
end Behavioral;
```

The input port **a** and output port **y** are defined as 1-bit STD_LOGIC signals. The statement **y <= not a;** assigns the inverted value of the input to the output.

2. Circuit and Simulation Results

The VHDL entity **not_gate** was imported into Logisim-evolution. An input pin was connected to port **a**, and an output pin was connected to port **y**. The circuit was simulated using the Questa VHDL simulator.

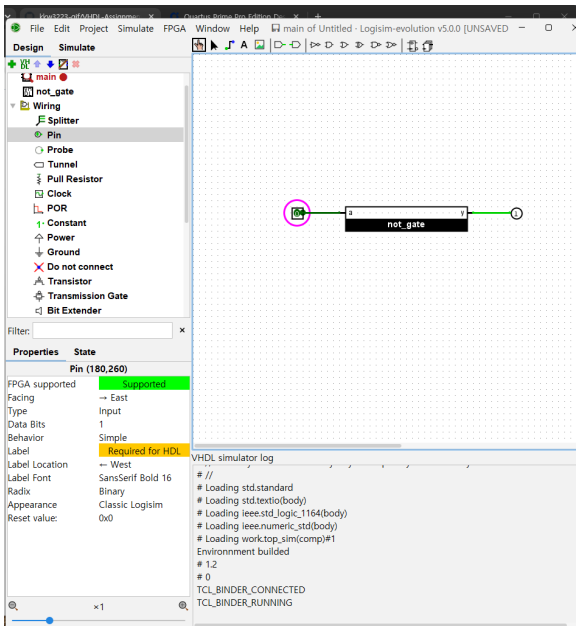


Figure 1. Simulation result for a = 0: y = 1

When the input **a** is 0, the output **y** becomes 1. This matches the expected behavior of a NOT gate.

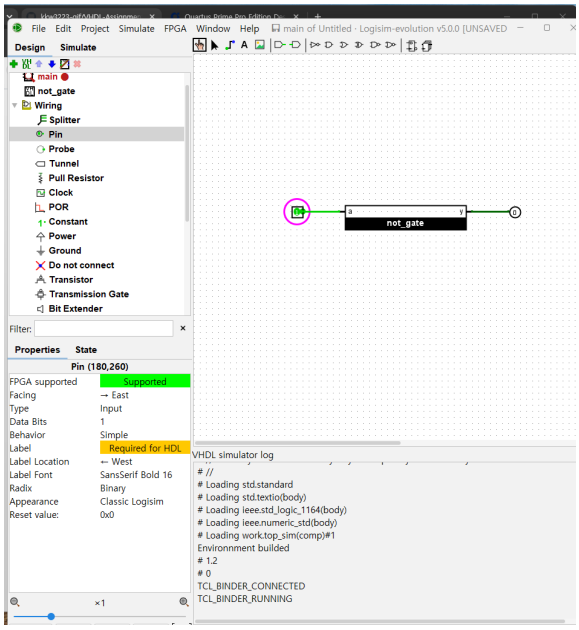


Figure 2. Simulation result for a = 1: y = 0

When the input **a** is 1, the output **y** becomes 0. The input value is correctly inverted.

3. Verification Results

Input a	Output y	Verification
0	1	PASS
1	0	PASS

Both test cases match the truth table of a NOT gate. Therefore, the VHDL implementation operates correctly in the Logisim-evolution and Questa simulation environment.

4. Conclusion

The NOT gate was successfully implemented in VHDL and verified through simulation. For an input of 0, the output was 1, and for an input of 1, the output was 0. These results confirm that the circuit performs the required logical inversion.